

classical nonlinear shock-capturing methods [19, 32, 42, 49]: the first is the uniqueness of approximation solution; and the second is that only a convergence rate of order $O(h^{1/2})$ can be proven in the energy norm for the linear element approximation, though numerical experiments demonstrating the optimal convergence rate $O(h)$.

In this paper, we develop a modified nonlinear DD finite element method for the convection-diffusion-reaction model (1.1), give results of existence and uniqueness for discrete solution, and derive a priori error estimates. Compared with [56], our main contributions lie in the following aspects:

- The considered model (1.1) is more general where the velocity β is not necessarily divergence free and the reaction coefficient σ is a function, while in [56] it has been assumed that $\nabla \cdot \beta = 0$ and σ is a constant;
- We modify the artificial diffusivity term to a proper dimensional scale, thus unify the two alternative variants of the artificial diffusion in [56] into one form. In the proof of the existence of approximation solutions, we do not assume the boundedness of the maximum norm of the gradient of functions in approximation space (see Remark 4.1). We further prove the uniqueness of approximation solution for the modified DD method under an assumption on small mesh size.
- We prove the optimal convergence rate $O(h)$ in the energy norm of approximation error plus a dissipative term for the modified DD method.

This paper is organized as follows. Section 2 gives notations and weak formulations of the model problem. Section 3 presents the modified DD scheme. Section 4 proves the existence and uniqueness of discrete solutions for the modified DD method, and also contains some preliminary results at the beginning of this section. Section 5 analyzes the optimal convergence order for the modified DD method. Finally, section 6 provides some numerical tests to support our theoretical results.

2 Notation and weak formulation

For any subdomain ω of Ω , denote $\|\cdot\|_{m,p,\omega}$ (resp. $|\cdot|_{m,p,\omega}$) the standard norm (resp. semi-norm) in the Sobolev space $W^{m,p}(\omega)$. In case of $p = 2$, $W^{m,2}(\omega) = H^m(\omega)$ ($H^0(\omega) = L^2(\omega)$) is the Hilbert space (with the inner product $(\cdot, \cdot)_\omega$) and is equipped with norm $\|\cdot\|_{m,\omega}$ (resp. $|\cdot|_{m,\omega}$); in case of $m = 0, p = \infty$, $W^{0,\infty}(\omega) = L^\infty(\omega)$ is the Lebesgue space with norm $\|\cdot\|_{0,\infty,\omega}$. We omit the symbol Ω in the notations above if $\omega = \Omega$, and also set

$$H_D^1(\Omega) = \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}.$$

Let $\mathcal{T}_h$ be a shape regular triangulation of Ω into triangles/tetrahedrons, and denote by h the mesh-size function with $h|_T = h_T := \text{diam}(T)$ for any $T \in \mathcal{T}_h$. Let $V_h \subset H^1(\Omega)$ be the usual linear element space, and ψ_T be the